REPRESENTATION THEORY PROJECT SUMMARY

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Introduction

Abstract. This project is an independent derivation of representation theory of matrix groups, using elementary results as a starting point. In particular, I assume chapters 1, 2, 3.1, 4, 6 from Linear Representations of Finite Groups by Serre [Ser77] and group-theoretic notions such as isomorphism theorems, Sylow groups and semi-direct products. First, I developed tools for working with induction and restriction of representations, including equivalence of different definitions of induction, criteria for irreducibility of induced representations, which led to rediscovering the standard notions of double cosets and normalizers. Second, I applied these tools, together with computational experiments, to reconstruct representations and character tables of matrix groups over finite fields and their subgroups. For instance, identifying the group of upper triangular two-by-two matrices as a semidirect product led to the study of iterated semidirect products of cyclic p-groups and to inverse problems concerning the recovery of group structure from representation-theoretic data, while analyzing normalizers in projective linear groups produced families of non-monomial characters.

Topic Selection. I selected my research topic independently. In one of the Representation Theory lectures at IISER, Dr. Manish hinted at the representation theory of matrix groups, which inspired me to pursue this challenge, reconnecting with the open-ended explorations from middle-school courses.

Logistics. I meet with Dr. Chandrasheel Bhagwat[1], an associate professor in the Mathematics Department at the Indian Institute of Science Education and Research, Pune[2], who specializes in representation theory and number theory, to present my progress and receive feedback, and learn the standard terminology for concepts I had independently rediscovered (double cosets, normalizers). I aim to consolidate this work into a sequence of questions that can guide a future student's exploration into this.

Role of Mentor. In my first meeting with Dr. Chandrasheel, I shared my interest, and we began with the simplest case: I wrote down the six elements of $\mathrm{GL}_2(\mathbb{F}_2)$, the group of 2×2 matrices over the field with two elements, and observed that these coincide with the standard representation of S_3. He noted that matrix groups are typically non-abelian, and since both groups have order 6, it was reasonable to conjecture that they are isomorphic. This continued: I presented questions to pursue each fortnight, and he introduced standard terminology (e.g., upper triangular matrices as a Borel subgroup) and emphasized rigor, urging me to prove my conjectures.

[1]Dr. Bhagwat's page: `https://www.iiserpune.ac.in/research/department/mathematics/people/faculty/regular-faculty/chandrasheel-bhagwat/239`.

[2]IISER: Indian Institute of Science Education and Research (IISER), Pune is a premier institute dedicated to research and teaching in the basic sciences, established in 2006 by the Ministry of Human Resource Development, India: `https://www.iiserpune.ac.in/`.

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Matrix Groups

Definition. Define the following subgroups of $\mathrm{GL}_2(\mathbb{F}_p)$, the group of 2×2 matrices with non-zero determinant and entries in the finite field $\mathbb{F}_p$. These definitions are readily generalizable for cases where $n > 2$.

- The *diagonal* subgroup: $\mathcal{D}_2(\mathbb{F}_p) = \{M \in \mathrm{GL}_2(\mathbb{F}_p) : M_{21} = 0 = M_{12}\}$.
- The so-called *Borel* subgroup: $\mathcal{B}_2(\mathbb{F}_p) = \{M \in \mathrm{GL}_2(\mathbb{F}_p) : M_{21} = 0\}$.
- The *unit-Borel* subgroup: $\mathcal{U}_2(\mathbb{F}_p) = \{M \in \mathcal{B}_2(\mathbb{F}_p) : M_{11} = 1 = M_{22}\}$.
- Then so-called *projective linear group*: $\mathrm{PGL}_2(\mathbb{F}_p) \cong \mathrm{GL}_2(\mathbb{F}_p)/\{\lambda I : \lambda \in \mathbb{F}_p^\times\}$.
- The *PDA (projective-diagonal-anti-diagonal)* subgroup of $\mathrm{PGL}_2(\mathbb{F}_p)$, $\mathrm{PDA}_2(\mathbb{F}_p)$, by
$$\mathrm{PDA}_2(\mathbb{F}_p) = N_{\mathrm{PGL}_2(\mathbb{F}_p)}\left(\mathcal{D}_2(\mathbb{F}_p) \cap \mathrm{PGL}_2(\mathbb{F}_p)\right).$$

Remark. These subgroups were crucial as I tackled the representation theory of the larger groups.

Observation. *We have the following structural simplifications. Let $\mathbb{F}_p^\times$ denote the group of integers modulo p under multiplication and $\mathbb{Z}/(p-1)\mathbb{Z}$ the group of integers modulo $p-1$ under addition:*

- $\mathcal{D}_2(\mathbb{F}_2) \cong \mathbb{F}_p^\times \cong \mathbb{Z}/(p-1)\mathbb{Z}$,
- $\mathcal{B}_2(\mathbb{F}_p) \cong \mathbb{Z}/p\mathbb{Z} \rtimes (\mathbb{Z}/(p-1)\mathbb{Z})^2$ *has presentation*
$$P = \left\langle r_1,\, r_2,\, r \;\middle|\; [r_1, r_2] = 1,\; r_1^{p-1} = 1,\; r_2^{p-1} = 1,\; r_1^{-1} r r_1 = r^{\alpha^{-1}},\; r_2^{-1} r r_2 = r^{\alpha} \right\rangle,$$
where α is a generator of $\mathbb{F}_p^\times$.
- *In general, note that $\mathcal{B}_n(\mathbb{F}_p) \cong \mathcal{U}_n(\mathbb{F}_p) \rtimes \mathcal{D}_n(\mathbb{F}_p)$.*

Proposition. *Recall that the group $\mathcal{D}_2(\mathbb{F}_p)$ has $(p-1)^2$ irreducible characters, given by χ_{k_1,k_2} where $0 \le k_1, k_2 < p-1$, defined by*
$$\chi_{k_1,k_2}(diag(d_1, d_2)) = \omega^{k_1 d_1 + k_2 d_2},$$
where ω is a $(p-1)^{th}$ root of unity. $\mathcal{B}_2(\mathbb{F}_p)$ has $(p-1)^2$ many one-dimensional characters, given by $\pi \circ \chi_{k_1,k_2}$ where $\pi : \mathcal{B}_2(\mathbb{F}_p) \to \mathcal{B}_2(\mathbb{F}_p)/[\mathcal{B}_2(\mathbb{F}_p), \mathcal{B}_2(\mathbb{F}_p)] = \mathcal{B}_2(\mathbb{F}_p)^{ab}$ is the canonical projection map. The functions
$$\boldsymbol{\chi}_{k_1,k_2} = Ind_{\mathcal{D}_2(\mathbb{F}_p)}^{\mathcal{B}_2(\mathbb{F}_p)} \chi_{k_1,k_2} - \pi \circ \chi_{k_1,k_2} \in \mathcal{C}(\mathcal{B}_2(\mathbb{F}_p)),$$
are irreducible characters of $\mathcal{B}_2(\mathbb{F}_p)$. Furthermore, $\boldsymbol{\chi}_{k_1,k_2} = \boldsymbol{\chi}_{k_1',k_2'}$ if and only if $k_1 + k_2 \equiv k_1' + k_2' \pmod{p-1}$, and so $\mathcal{B}_2(\mathbb{F}_p)$ has $p-1$ many $(p-1)^2-$dimensional irreducible characters.

Remark. By the sum-of-squares formula, these are all the representations of $\mathcal{B}_2(\mathbb{F}_p)$.

Corollary. *The group $\mathcal{B}_2(\mathbb{F}_p) \cap PGL_2(\mathbb{F}_p)$, of order $p(p-1)$, has $p-1$ one-dimensional representations, and for any k_1, k_2,*
$$C'_{k_1,k_2} = Res_{\mathcal{B}_2(\mathbb{F}_p) \cap PGL_2(\mathbb{F}_p)}^{\mathcal{B}_2(\mathbb{F}_p)} \boldsymbol{\chi}_{k_1,k_2}$$
is the unique $(p-1)$-dimensional representation of $\mathcal{B}_2(\mathbb{F}_p) \cap PGL_2(\mathbb{F}_p)$.

Remark. By the sum-of-squares formula, these are all the characters of $\mathcal{B}_2(\mathbb{F}_p) \cap \mathrm{PGL}_2(\mathbb{F}_p)$.

Remark. This was the only subgroup I was armed with entering $\mathrm{PGL}_2(\mathbb{F}_p)$. Later, $\mathrm{PDA}_2(\mathbb{F}_p)$ would prove to be useful.

Lemma. *We have that $[PGL_2(\mathbb{F}_p), PGL_2(\mathbb{F}_p)] \cong \mathbb{Z}/2\mathbb{Z}$. In particular, there are 2 one-dimensional representations of $PGL_2(\mathbb{F}_p)$: $\mathbf{1}_{PGL_2(\mathbb{F}_p)}$ and ε, where $\varepsilon(M) = \left(\frac{\det M}{p}\right)$, and $\left(\frac{\cdot}{p}\right)$ is the Legendre symbol.*

Proposition. *Define $I_k \in \mathcal{C}(\mathcal{B}_2(\mathbb{F}_p) \cap PGL_2(\mathbb{F}_p))$ for $0 \le k < p-1$ by*
$$I_k\left(\begin{bmatrix} 1 & b \\ 0 & d \end{bmatrix}\right) = \omega^{k\phi(d)},$$
where where ϕ is an isomorphism from $\mathbb{F}_p^\times$ to $\mathbb{Z}/(p-1)\mathbb{Z}$. Then, the I_k are the one-dimensional characters of $\mathcal{B}_2(\mathbb{F}_p) \cap PGL_2(\mathbb{F}_p)$. Furthermore,
$$Ind_{PGL_2(\mathbb{F}_p) \cap \mathcal{B}_2(\mathbb{F}_p)}^{PGL_2(\mathbb{F}_p)} I_k$$
is an irreducible character of dimension $p+1$ of $PGL_2(\mathbb{F}_p)$ for $k \neq 0$ and $k \neq (p-1)/2$. Indeed, we have
$$Ind_{PGL_2(\mathbb{F}_p) \cap \mathcal{B}_2(\mathbb{F}_p)}^{PGL_2(\mathbb{F}_p)} I_{(p-1)/2} = \varepsilon \oplus \varepsilon',$$
where ε is the non-trivial one-dimensional character of $PGL_2(\mathbb{F}_p)$ and ε' is an irreducible character of $PGL_2(\mathbb{F}_p)$ of dimension p.

Remark. Notice the image of the homomorphism $\varphi : \mathrm{PGL}_2(\mathbb{F}_p)^{\mathrm{ab}} \to (\mathcal{B}_2(\mathbb{F}_p) \cap \mathrm{PGL}_2(\mathbb{F}_p))^{\mathrm{ab}}$.

Example. $\mathcal{B}_2(\mathbb{F}_3)$ has 4 one-dimensional characters and 2 2−dimensional characters.

	$\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$	$\begin{pmatrix}-1 & 0\\ 0 & -1\end{pmatrix}$	$\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}$	$\begin{pmatrix}-1 & 1\\ 0 & -1\end{pmatrix}$	$\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$	$\begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}$
$\chi_{(0,0)}$	1	1	1	1	1	1
$\chi_{(1,0)}$	1	−1	1	−1	1	−1
$\chi_{(0,1)}$	1	−1	1	−1	−1	1
$\chi_{(1,1)}$	1	1	1	1	−1	−1
$\boldsymbol{\chi}_0$	2	2	−1	−1	0	0
$\boldsymbol{\chi}_1$	2	−2	−1	1	0	0

Question. *Does there exist a group G such that is not isomorphic to $\mathcal{B}_2(\mathbb{F}_3)$, but has the same character table? Alternatively, let G_d be a finite group such that G_d has d^2 many one-dimensional representations and d many $d-$dimensional representations only. What can one say about G?*

Lemma. *We have the double coset decomposition*

$$PDA_2(\mathbb{F}_p)\backslash PGL_2(\mathbb{F}_p)/PDA_2(\mathbb{F}_p) \;=\; PDA_2(\mathbb{F}_p) \;\sqcup\; \bigsqcup_{\mu\in S} PDA_2(\mathbb{F}_p)\, r_\mu\, PDA_2(\mathbb{F}_p) \;\sqcup\; PDA_2(\mathbb{F}_p)\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix} PDA_2(\mathbb{F}_p),$$

where S contains -1 and is such that $x \in S \implies x^{-1} \notin S$. Further, r_μ is such that $(r_\mu)_{i,j} = 1$ for $(i,j) \neq (2,2)$ and $(r_\mu)_{(2,2)} = \mu$. Sizes of the double cosets: first $2(p-1)$, $PDA_2(\mathbb{F}_p) r_\mu PDA_2(\mathbb{F}_p)$ $2(p-1)^2$, last $4(p-1)^2$. Indeed, note that

$$2\cdot(p-1)^2\cdot\frac{p-2}{2} + 2(p-1) + 4(p-1)^2 = p(p+1)(p-1) = |PGL_2(\mathbb{F}_p)|.$$

Remark. This is decomposition is used to compute the $\mathrm{Ind}_{\mathrm{PDA}_2(\mathbb{F}_p)}^{\mathrm{PGL}_2(\mathbb{F}_p)}\kappa$, where κ is one of the four one-dimensional characters of $\mathrm{PDA}_2(\mathbb{F}_p)$. The representations induced from $\mathcal{B}_2(\mathbb{F}_p) \cap \mathrm{PGL}_2(\mathbb{F}_p)$ seem to be constituents, which may possibly leave a direct sum of characters of the same dimension.

Induced Representation Tools

Remark. One of the aspects of induced representations that I worked with was proving the equivalence of the various definitions (via function spaces, tensor products, and actions on coset spaces). In particular, let H be a subgroup of G, and fix a set of left coset representatives $r_1, \ldots, r_{[G:H]}$. For $g \in G$, define a permutation $\sigma_g\colon \{1,\ldots,[G:H]\} \to \{1,\ldots,[G:H]\}$ by requiring that $r_{\sigma_g(i)}^{-1} g r_i \in H$, and define $h(g,i) = r_{\sigma_g(i)}^{-1} g r_i \in H$. Let W be a $K[H]$-module. After experimentation, I defined a $K[G]$-module structure on the abelian group $\bigoplus_{i=1}^{[G:H]} W_i$, where each $W_i \cong W$, by

$$g\cdot\sum_{i=1}^{[G:H]} w_i = \sum_{i=1}^{[G:H]} h\big(g, \sigma_g^{-1}(i)\big)\, w_{\sigma_g^{-1}(i)},$$

for $\sum_i w_i \in \bigoplus_{i=1}^{[G:H]} W_i$. This action is compatible with the map

$$\phi\colon K[G]\otimes_{K[H]} W \longrightarrow \bigoplus_{i=1}^{[G:H]} W_i$$

defined by $\phi(g\otimes w) = \sum_{i=1}^{[G:H]} f(i)\, w$, with $f(i) = \delta_{i,\sigma_g(e)}$ where e denotes the index corresponding to the identity coset. The key reason this $K[G]-$module structure worked out (associativity) because of the following identity satisfied by h for all $g_1, g_2 \in G$ and $1 \le i \le [G:H]$:

$$h(g_1 g_2, i) = h(g_1, \sigma_{g_2}(i))h(g_2, i).$$

Code to assist with computational experiments

Throughout this project, I had to write code to verify some of my conjectures. For instance, the following Python code generates the elements of the group $\mathrm{GL}_n(\mathbb{F}_p)$.

```
def generate_matrices(n, p):
    vectors = dynamic_loops(n, p)
    vectors = [red_sym_ma(v, p) for v in vectors]
    del vectors[0]
    matrices = []
    count = 0
    matrix = np.zeros((n, n))
    matrices.append(matrix)

    while count < n:
        for m in matrices:
            index = get_index(m, matrices, p)
            l = []
            for v in array_list_difference(
                    vectors,
                    multiples_matrix(p, m, count)):
                m_new = m.copy()
                m_new[:, count] = v
                l.append(m_new)
            matrices[index] = l

        count += 1
        matrices = [i for s in matrices for i in s]

    return [m.astype(int) for m in matrices]
```

References

[Ser77] Jean-Pierre Serre. *Linear Representations of Finite Groups*, volume 42 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1977. Chapters 1, 2, 3.1, 4, and 6.

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